# Features and Distractors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

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WORKBOOK · 1 OF 21

## Recall and choose your starting point

> Record your starting explanation for Features and Distractors before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Operational Labels

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

### Debugging Strategies

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

### Branching Logic

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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WORKBOOK · 2 OF 21

## Practise with fading help

> Complete the supported Features and Distractors trace and record the first step you can explain without help.

Use W2: Prove that one measurement is insufficient for **Feature**s and useful distinctions. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

**The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 21

## Individual reasoning check

> Check the proposed Features and Distractors explanation against the supplied facts before discussing it.

Use the worked case for **Feature**s and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

WORKBOOK · 4 OF 21

## Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

**Independent case: Changed independent case**

**Training** covers show space books only on yellow backgrounds and animal books only on blue backgrounds. Design two controlled comparisons to test whether a classifier uses the intended subject.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

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WORKBOOK · 5 OF 21

## Your learning target

> Connect the Features and Distractors target to a trace and a justified conclusion.

All animal books in **training** have green covers. How could you test whether a recommender is using subject or cover colour? Supply two diagnostic test descriptions.

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WORKBOOK · 6 OF 21

## Solve the given case

> Solve the supplied Features and Distractors case and show the deciding evidence.

Target: includes animal characters. Fields: animal tag, cover colour and shelf number.

Choose useful **evidence** and possible distractions.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

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WORKBOOK · 7 OF 21

## Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Predict whether a book is short using page count and cover colour. Which **feature** directly supports the defined label? Give a counterexample to colour.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

### Week 9 Test and reflect

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WORKBOOK · 8 OF 21

## Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

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## Project checkpoint

> Justify what Features and Distractors adds to Transparent Book Recommender and keep the evidence.

Use explainable traits that connect to the reader's stated preferences.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

## Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

## Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

All animal books in **training** have green covers. How could you test whether a recommender is using subject or cover colour? Supply two diagnostic test descriptions.

**My independently reasoned answer**

Record support used: none, hint or teacher explanation.

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WORKBOOK · 12 OF 21

## W1 Separate inputs from the target

> Separate inputs from the target. Apply the supplied Features and Distractors facts and keep your reasoning.

For the fixed-orientation card task, fits requires length at most 12 cm and width at most 8 cm, including equality.

Choose the measurements needed as **input**s from length, width, colour, card number and the final inspection sticker. State the **target label** and write the **feature record** for a 12 cm by 8 cm red card using a defined order.

> **My reasoning and evidence**

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WORKBOOK · 13 OF 21

## W2 Prove that one measurement is insufficient

> Prove that one measurement is insufficient. Apply the supplied Features and Distractors facts and keep your reasoning.

Invent two cards with the same width but different correct labels under the card guide. Give both lengths and widths and justify the labels. Explain why a system receiving only width cannot always label both correctly.

**My reasoning and evidence**

WORKBOOK · 14 OF 21

## W3 Challenge an apparent colour shortcut

> Challenge an apparent colour shortcut. Apply the supplied Features and Distractors facts and keep your reasoning.

A practice set contains three blue cards that fit and three red cards that are too large. A helper uses blue means fits and red means too large. Create a new card that this helper mislabels.

Explain why the perfect practice score did not establish that colour was a reliable **feature** for this task.

**My reasoning and evidence**

WORKBOOK · 15 OF 21

## W4 Design a record for a new task

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Design a record for a new task. Apply the

> supplied Features and Distractors facts and keep your reasoning.

A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed. Before checking, you can record height, thickness and pattern name.

A pass sticker is added afterward. Choose **input**s with units. Construct two tiles with the same height but different labels.

Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an **input** before checking.

> **My reasoning and evidence**

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WORKBOOK · 16 OF 21

## W5 Notice when the task changes

> Notice when the task changes. Apply the supplied Features and Distractors facts and keep your reasoning.

The delivery station now sorts cards into red and blue bins instead of checking dimensions. Which previously unneeded **feature** becomes useful? Explain why length and width alone may now be insufficient.

Give two cards that support your explanation.

**My reasoning and evidence**

WORKBOOK · 17 OF 21

## Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

| Evidence | My record |
| --- | --- |
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

## Pictorial case practice: Features and Distractors

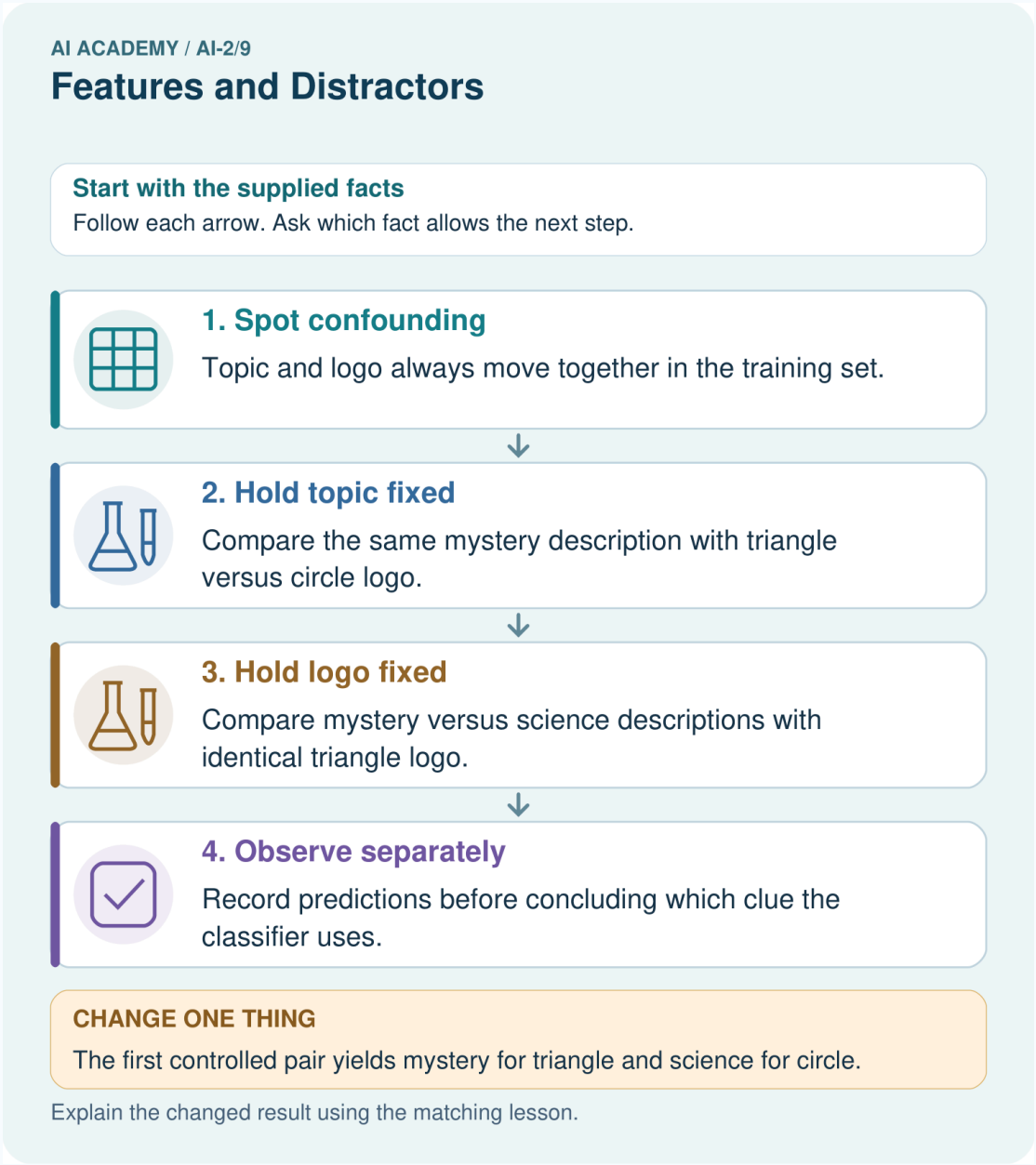
> A fictional book-topic classifier sees mystery books with triangular logos and science books

> with circular logos. Explain the labelled step before checking the changed condition.

A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.

**Training** contains no mystery-circle or science-triangle covers.

Goal is to identify topic from actual subject information, not publisher logo. New controlled test **output**s have not been supplied.




Follow the labelled picture with the matching explanation. Open the vector diagram

Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

**My picture-based explanation and changed-case prediction**

### Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Test recommender **feature**s for **shortcut**s, keeping **relevant** subject information separate from superficial catalogue styling.

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## Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Test recommender **feature**s for **shortcut**s, keeping **relevant** subject information separate from superficial catalogue styling.

| Evidence to keep | My record |
|---|---|
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

## Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Record **prediction**s before concluding which clue the classifier uses.

1. Hold topic fixed

2. Hold logo fixed

3. Observe separately

4. Spot confounding

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Observe separately Record **prediction**s before concluding which clue the classifier uses.

Either subject information or logo could explain success on the old combinations. Controlled pairs separate their **prediction**s by changing one aspect at a time.

Designing the pair is not equivalent to observing the model’s response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

### Practice 2: explain the deciding evidence

#### Changed case · predict first

Changed-case practice: The first controlled pair yields mystery for triangle and science for circle. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. That pair is **evidence** of unwanted logo sensitivity.

**My selected conclusion, deciding fact and explanation**

#### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: That pair is **evidence** of unwanted logo sensitivity. The subject description stayed mystery; only the ir**relevant** logo changed.

### Repair a missing building block

Review Operational Labels (ai-2/8). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Why can we not resolve the disagreement merely by taking a majority of reviewers?.

Return to this week and explain what you changed.

Review Debugging Strategies (ai-2/6). Goal: Design revealing tests for a function. Short practice: Which test most directly isolates the handling of loose books, and what should it produce?.

Return to this week and explain what you changed.

Review Branching Logic (ai-2/3). Goal: Trace the selected branch in an if elif else chain. Short practice: Give expected labels for –1,0,10,11 and identify the equality boundary for Short..

Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

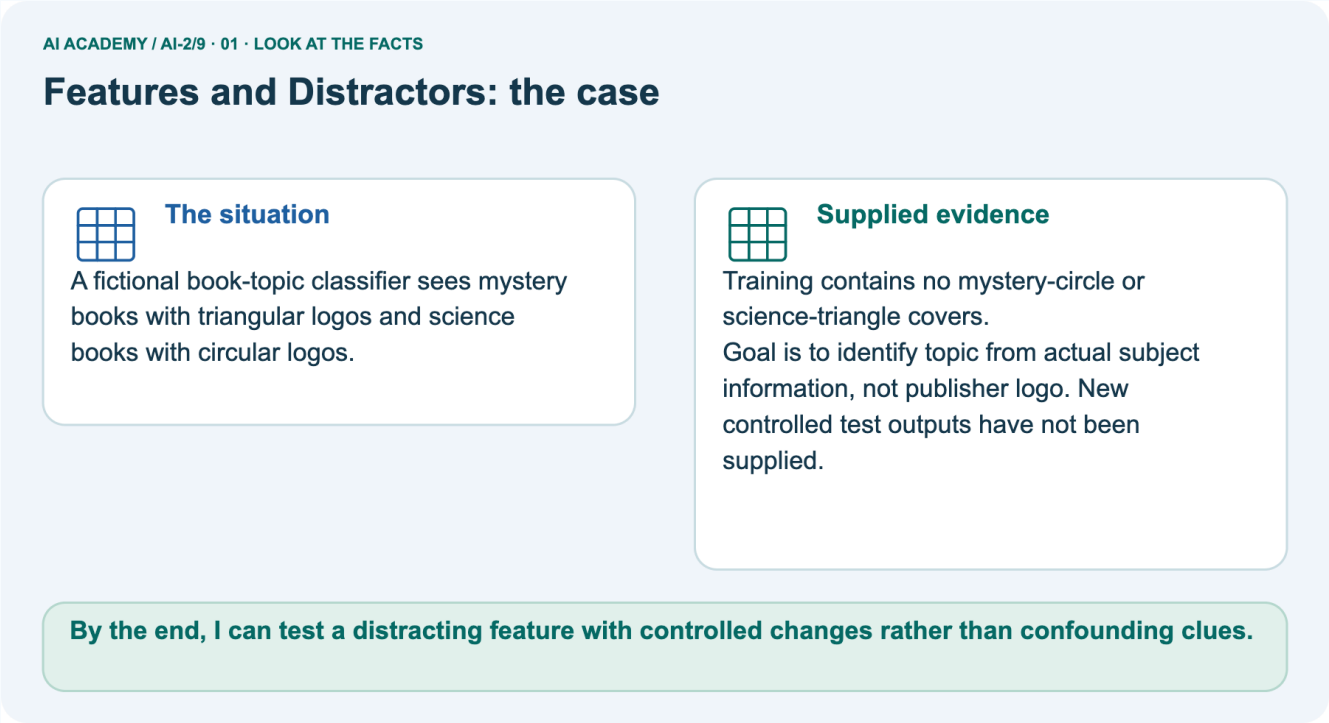
Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

# Visual concept journey

> Explain each Features and Distractors picture from its labels, then record a relevant limit.

### 01. Features and Distractors: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

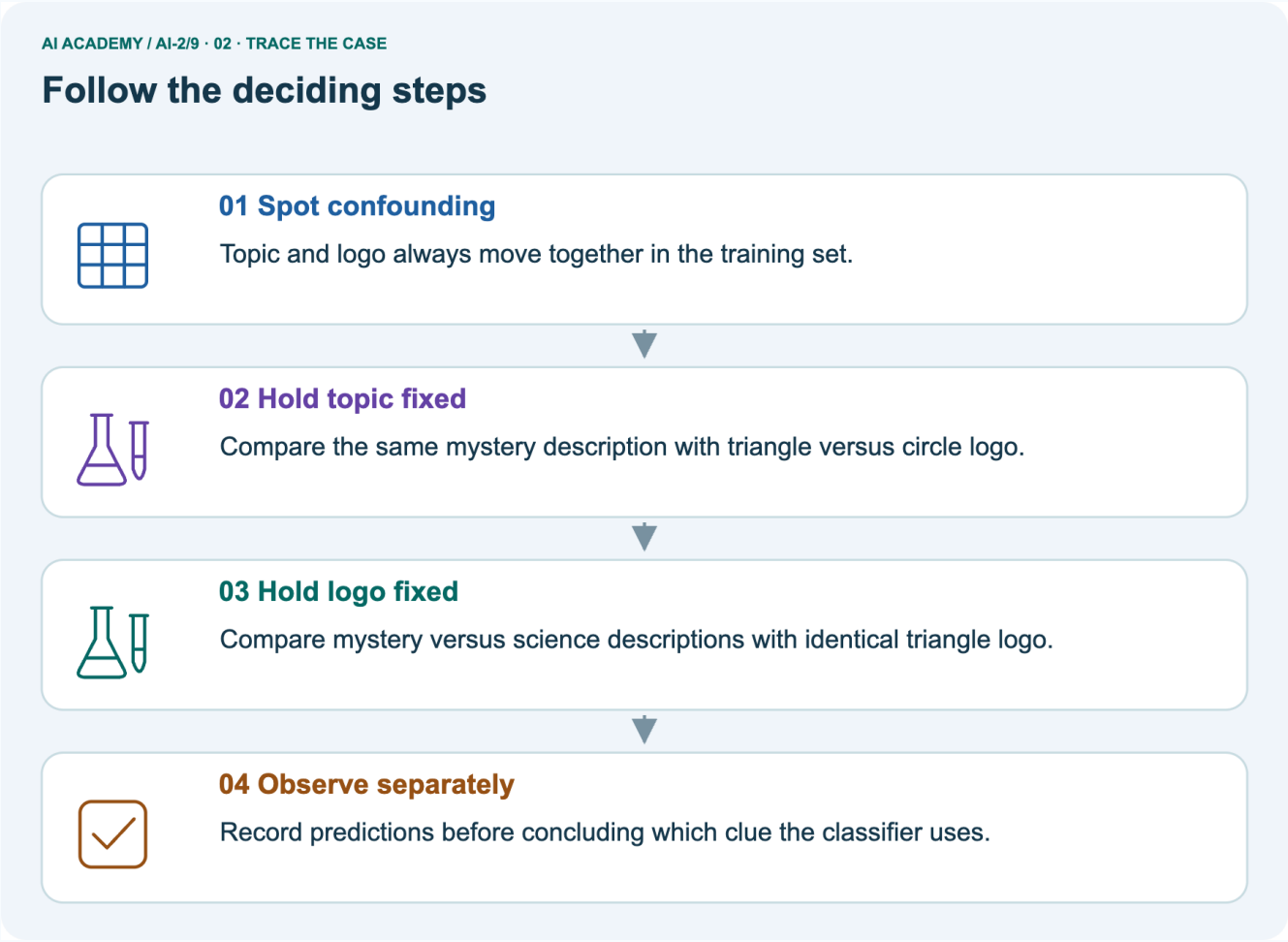
A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos. Supplied facts: **Training** contains no mystery-circle or science-triangle covers.

Goal is to identify topic from actual subject information, not publisher logo. New controlled test **output**s have not been supplied.

> Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



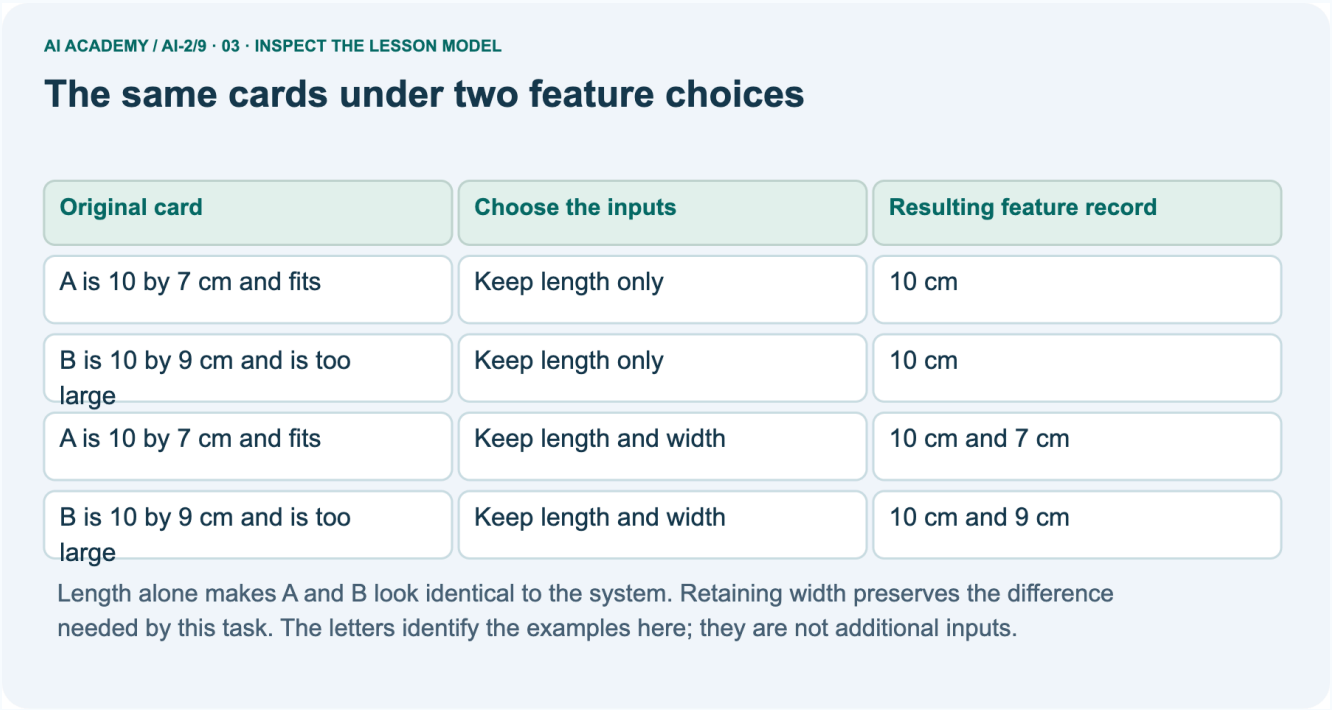

Follow the labelled picture with the matching explanation. Open the vector diagram

Spot confounding: Topic and logo always move together in the **training** set. → Hold topic fixed: Compare the same mystery description with triangle versus circle logo. → Hold logo fixed: Compare mystery versus science descriptions with identical triangle logo. → Observe separately: Record **prediction**s before concluding which clue the classifier uses.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

### 03. The same cards under two feature choices




Follow the labelled picture with the matching explanation. Open the vector diagram

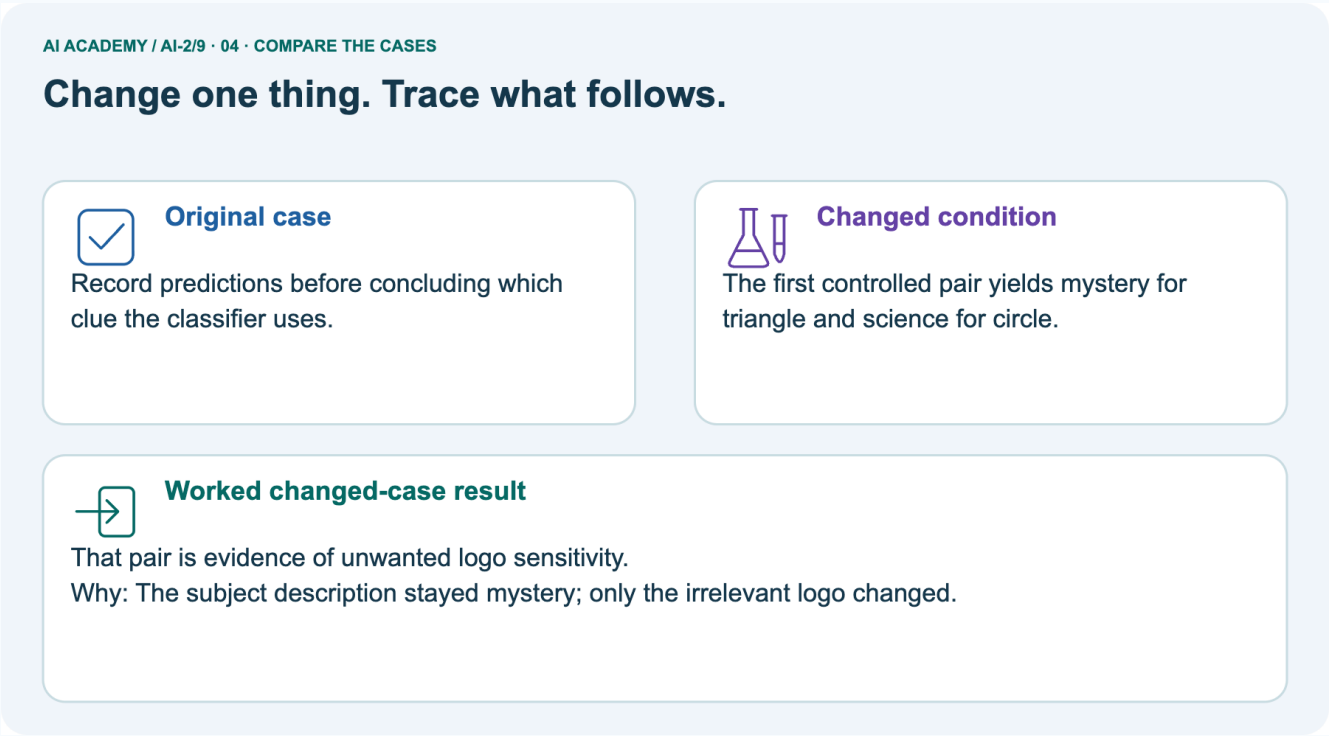
The same cards under two **feature** choices. A is 10 by 7 cm and fits | Keep length only | 10 cm. B is 10 by 9 cm and is too large | Keep length only | 10 cm. A is 10 by 7 cm and fits | Keep length and width | 10 cm and 7 cm.

B is 10 by 9 cm and is too large | Keep length and width | 10 cm and 9 cm

Read the labels and values. Explain the relationship the model shows.

**My picture explanation, deciding evidence and one limit**

### 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

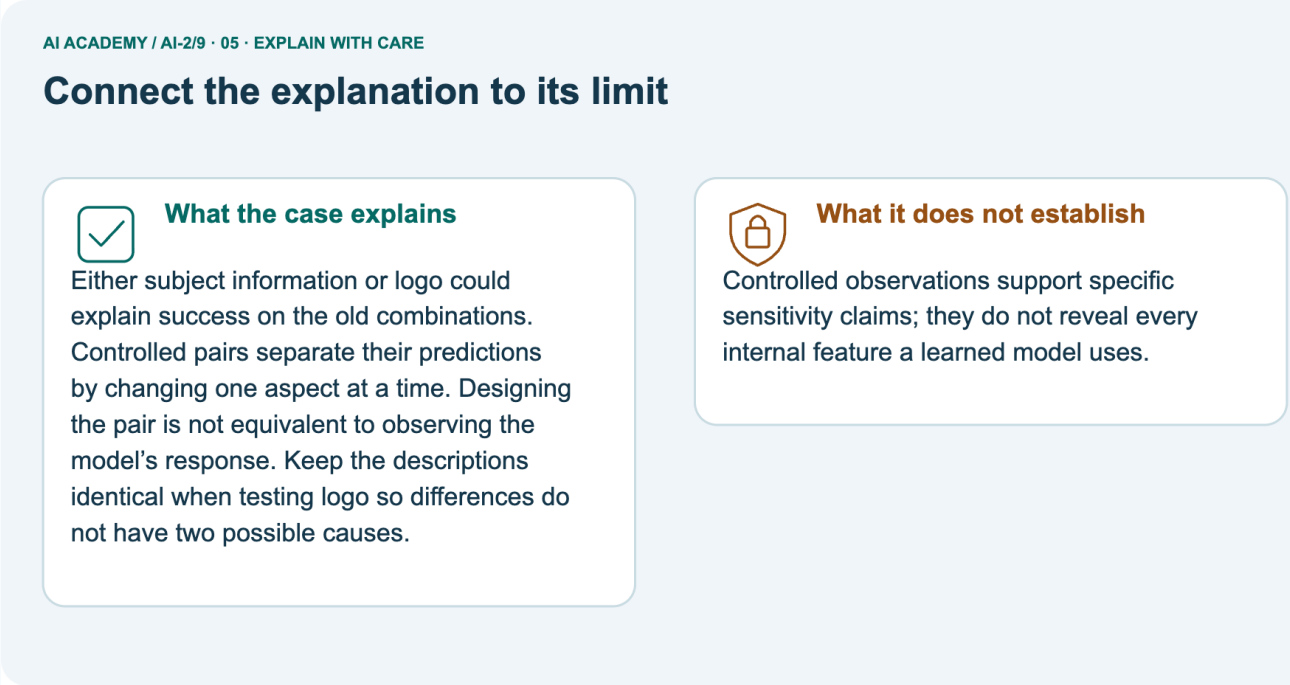
Original case: Record **prediction**s before concluding which clue the classifier uses. Changed condition: The first controlled pair yields mystery for triangle and science for circle.

Expected result: That pair is **evidence** of unwanted logo sensitivity. Reason: The subject description stayed mystery; only the ir**relevant** logo changed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

**My picture explanation, deciding evidence and one limit**

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Either subject information or logo could explain success on the old combinations. Controlled pairs separate their **prediction**s by changing one aspect at a time.

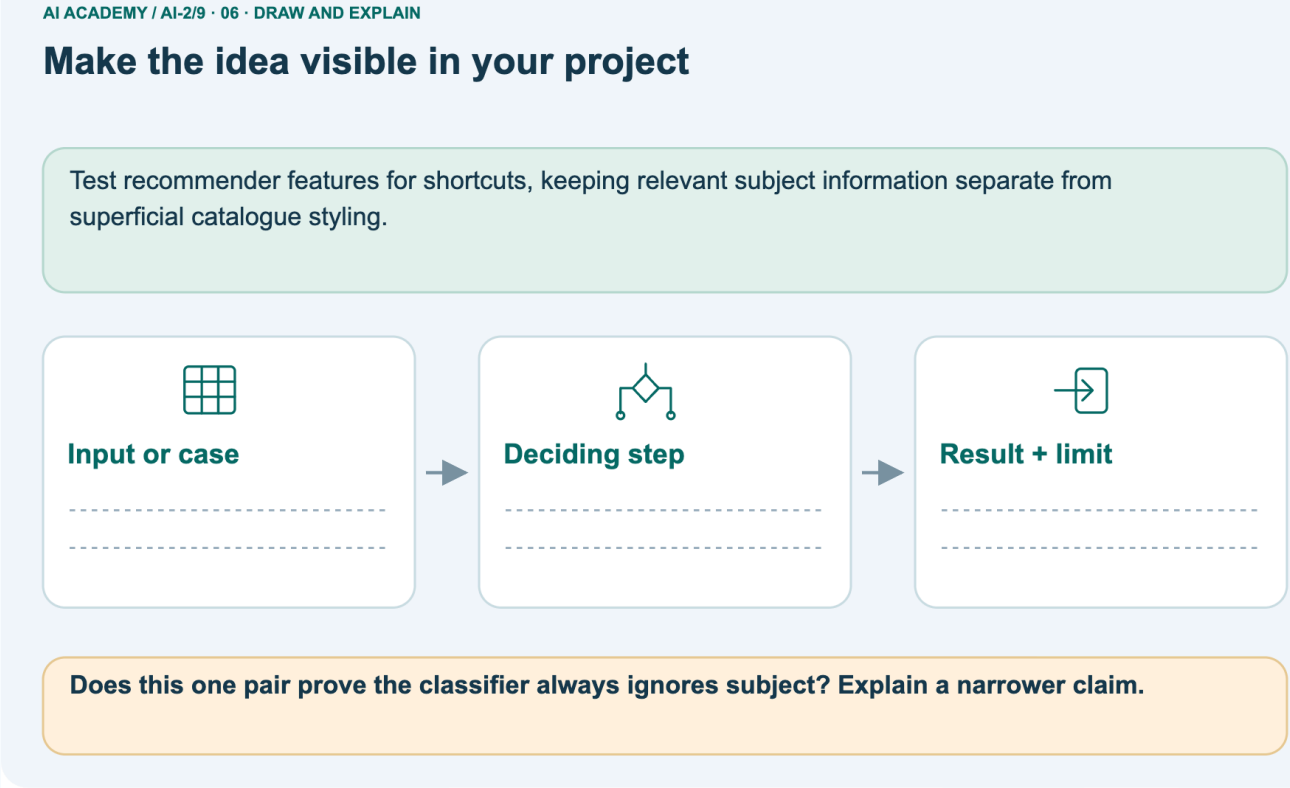
Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

Limit: Controlled **observation**s support specific sensitivity claims; they do not reveal every internal **feature** a **learned model** uses.

> Explain the case, then point to a limit on the claim.

**My picture explanation, deciding evidence and one limit**

### 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Test recommender **feature**s for **shortcut**s, keeping **relevant** subject information separate from superficial catalogue styling. Draw your **input**, deciding step, result and limit.

Supported question: Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

> Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.